

Muhammad Hassan Askari

Junior Software Engineer

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PROFESSIONAL SUMMARY

Detail-oriented recent BS Computer Science graduate with a strong foundation in full-stack software engineering and DevOps practices. Experienced in designing scalable backend architectures, developing cross-platform applications, and automating deployment workflows. Proficient in modern frameworks (ASP.NET Core, Laravel, React, Flutter) and passionate about leveraging cloud technologies (AWS, Docker) and CI/CD pipelines to build, deploy, and maintain reliable, high-performance software solutions.

TECHNICAL SKILLS

- **Programming Languages:** C#, Python, PHP, C++, Dart, JavaScript, SQL.
- **AI & Machine Learning:** RAG, LLM Integration, Semantic Search, Vector Embeddings, Prompt Engineering.
- **Cloud & DevOps:** AWS (EC2), Docker, Kubernetes, Jenkins, Git, GitHub Actions, NGINX.
- **Backend & Web Frameworks:** .NET, ASP.NET Core & Web API, Laravel, React.js, REST API Design.
- **Mobile & Real-Time:** Flutter, SignalR, Firebase Cloud Messaging (FCM).
- **Databases:** PostgreSQL, MySQL, Oracle Database, Firebase Firestore.

PROFESSIONAL EXPERIENCE

Pakistan Telecommunication Authority (PTA) | *ICT Intern* Jul 2025 – Sep 2025

- Developed a robust backend using ASP.NET Web API for an E-Diary system, successfully digitizing internal documentation and workflow processes.
- Designed and implemented relational database schemas optimizing for efficient data storage and rapid retrieval.
- Built the frontend application using Flutter, ensuring seamless integration with the backend REST APIs and a highly responsive user interface.
- Collaborated within an Agile development environment, actively participating in sprint meetings and iterative progress reviews.

HBL Microfinance Bank Ltd. | *Software Quality Assurance Intern* Jul 2024 – Sep 2024

- Collaborated closely with the software development team within an Agile framework to identify defects and ensure timely issue resolution across web and Android applications.
- Programmed VBA macros and utilized Excel to automate repetitive QA tasks, significantly improving testing efficiency and analytical reporting.
- Gained comprehensive exposure to the Software Development Life Cycle (SDLC) by evaluating system integration and access control mechanisms.

EDUCATION

BS Computer Science Sep 2022 – Jun 2026
COMSATS University Islamabad, Wah Campus CGPA: 3.65 / 4.00

KEY PROJECTS

AI Email Copilot

Aug 2026

- Developed an AI-powered Gmail assistant using FastAPI, React, TypeScript, PostgreSQL, and pgvector to enable intelligent email search, analysis, and context-aware response generation.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using semantic search and all-MiniLM-L6-v2 embeddings to retrieve relevant historical email context before generating AI responses.
- Integrated Gmail API and OAuth 2.0 for secure email synchronization and multi-account access, with background processing through APScheduler.
- Built an AI inbox chat interface that answers questions using retrieved email sources and generates context-aware email drafts using LLM inference.

OpenHouse & Job Fair Portal (Final Year Project)

Sep 2025 – Jun 2026

- Architected a full-stack recruitment platform utilizing React, Flutter, and ASP.NET Core 8 Web API, featuring real-time communication via SignalR and Firebase.
- Successfully beta tested the application during the Spring 2026 job fair at CUI Wah Campus, facilitating interaction for over 150 students and 20+ recruiting companies.
- Developed a conflict-free auto-scheduling algorithm to efficiently allocate hundreds of interview slots under real-world time constraints.
- **DevOps Implementation:** Deployed and maintained production infrastructure on AWS EC2 and the local university server. Containerized the application using Docker and configured NGINX as a reverse proxy for multiple role-based subdomains.

LaneGuard – Monocular Lane Perception Benchmark

2026

- Developed a computer-vision evaluation pipeline comparing classical OpenCV-based lane detection with lightweight YOLOv8n-seg and YOLO26n-seg segmentation models.
- Built a reproducible evaluation workflow on the TuSimple dataset, including lane-boundary extraction, lane-center estimation, failure analysis, and heuristic lane-departure warning.
- Evaluated the approaches on 2,782 held-out frames using boundary accuracy, lane-center error, failure rate, and inference latency on an NVIDIA Tesla T4.

Event Reminder App

Apr 2025 – May 2025

- Built a cross-platform mobile application using Flutter for event scheduling and automated reminders.
- Integrated Firebase Authentication and Firestore for secure user management and real-time data synchronization.

Real Estate Management System

Apr 2025 – May 2025

- Developed a dynamic property management web application utilizing Laravel and MySQL, enabling users to list, search, and manage real estate data.
- Implemented secure user authentication, dynamic listings, and full CRUD operations for comprehensive property data management.

CERTIFICATIONS & LEADERSHIP

Software QA & Test Automation Engineering Specialization

Jul 2026

Completed 5 courses covering Selenium, Agile and automated testing, Cypress, JMeter, xUnit, and API validation with Postman.

Google Cybersecurity Professional Certificate

Sep 2023

Mastered foundational security principles, threat modeling, Linux system administration, and Python automation workflows.

Mobile App Development Competition: Competitor in the 21st Episode of VisioSpark

Nov 2025

Student Startup Business Society (SSBS): Head of Operations / Advisor

Sep 2024 – Jun 2026

Demonstrated leadership, team coordination, and event management.